

Functional Analysis

2026-09-26

The foundational knowledge relating to Measure & Ergodic Theory, that has not been covered elsewhere, was provided by Jamneshan and Kreidler (2025).

Definition 0.1. For a Hilbert space H ,

1. A *linear isometry* is a linear map $U : H \rightarrow H$ such that $\|Uf\| = \|f\|$ for all $f \in H$ (and thus injective).
2. A linear isometry, U , is a *unitary operator* if it is surjective (and thus bijective).
3. We write $\mathcal{U}(H)$ for the set of all unitary operators $U : H \rightarrow H$.
4. For a group Γ , we call a group homomorphism $U : \Gamma \rightarrow \mathcal{U}(H)$, $\gamma \mapsto U_\gamma$ a *unitary representation* of Γ on H .

Definition 0.2. Let $J : (Y, S) \rightarrow (X, T)$ be an extension of measure-preserving systems. Then $E = U_J(L^2(Y))$ is an *invariant Markov sub-lattice* of $L^2(X)$, i.e.,

1. E is a closed linear subspace of $L^2(X)$,
2. $1 \in E$,
3. $|f|, \operatorname{Re}(f), \operatorname{Im}(f) \in E$ for every $f \in E$, and
4. $U_{T_\gamma} f \in E$ for every $f \in E$ and $\gamma \in \Gamma$.

Definition 0.3. Let H be a Hilbert space. $\mathcal{L}(H)$ is the space of all bounded linear operators from H to H .¹

A family $\mathcal{S} \subseteq \mathcal{L}(H)$ is a *semigroup (of operators)* if $UV \in \mathcal{S}$ for all $U, V \in \mathcal{S}$. It is a *contraction semigroup* if, in addition, $\|U\| \leq 1$ for all $U \in \mathcal{S}$.

We call

$$\operatorname{fix}(\mathcal{S}) := \bigcap_{U \in \mathcal{S}} \operatorname{fix}(U) = \{f \in H \mid Uf = f \text{ for every } U \in \mathcal{S}\}$$

the *fixed space* of $\mathcal{S}$.²³

¹Note, $\mathcal{U}(H) \subseteq \mathcal{L}(H)$ and $\|U\| = 1$ when $U \in \mathcal{U}(H)$, so $\mathcal{U}(H)$ is a contraction semigroup.

² $\operatorname{fix}(\mathcal{S}) \subseteq H_{\text{ds}}$.

³Meeting Notes: 0-dimensional. Next most simple case would be scaling (1-dimensional), and then rotations on a plane (2-dimensional).

The *closed convex hull* $\overline{\text{co}} A$ is the closure of the set of all convex combinations of elements of A .

The *closed linear hull* $\overline{\text{lin}} A$ is the closure of the set of all linear combinations of elements of A .

Definition 0.4. We write $\mathbb{T} := \{z \in \mathbb{C} \mid |z| = 1\}$ for the multiplicative group of complex numbers of modulus one, and call this the *torus*.

Definition 0.5. Let $U : \Gamma \rightarrow \mathcal{U}(H)$ be a unitary representation of a discrete abelian amenable group Γ .

A group homomorphism $\chi : \Gamma \rightarrow \mathbb{T}$, where $\mathbb{T} = \{z \in \mathbb{C} \mid |z| = 1\}$, is called a *character*. The *dual group* Γ^* of Γ is the set of all such characters equipped with the multiplication given by $(\chi_1 \chi_2)(\gamma) := \chi_1(\gamma) \chi_2(\gamma)$ for $\gamma \in \Gamma$ and $\chi_1, \chi_2 \in \Gamma^*$.⁴

Definition 0.6. Let

$$M_f := \text{lin}\{U_\gamma f \mid \gamma \in \Gamma\}.$$

We call a subset $M \subseteq H$ an *invariant finite-dimensional subspace* if M_f is finite-dimensional and $U_\gamma f \in M$ for all $f \in M$ and $\gamma \in \Gamma$.

The closure

$$H_{\text{ds}} = \overline{\bigcup \{M \subseteq H \mid M \text{ invariant finite-dimensional subspace}\}} \subseteq H$$

is called the *discrete spectrum part* of U .⁵

Definition 0.7 (Jamneshan and Kreidler, 2025, Definition 5.1.8). Let $U : \Gamma \rightarrow \mathcal{U}(H)$ be a unitary representation of an abelian group Γ . For a character $\chi \in \Gamma^*$, we call

$$\begin{aligned} \ker(\chi - U) &:= \{f \in H \mid U_\gamma f = \chi(\gamma)f \text{ for every } \gamma \in \Gamma\} \\ &\quad \bigcap_{\gamma \in \Gamma} \ker(\chi(\gamma)\text{Id}_H - U_\gamma) \end{aligned}$$

the *eigenspace* of U with respect to χ , and elements $f \in \ker(\chi - U) \setminus \{0\}$ are called *eigenvectors* with respect to χ . The set $\sigma_p(U) := \{\chi \in \Gamma^* \mid \ker(\chi - U) \neq \{0\}\}$ is the *point spectrum* of U and its elements are called the *eigenvalues* of U .

Proposition 0.1 (cf. Jamneshan and Kreidler, 2025, Proposition 5.1.10). *The eigenspaces $\ker(\chi - U)$ for $\chi \in \Gamma^*$ are pairwise orthogonal. For $M \subseteq H$, then the following are equivalent:*

1. *M is an irreducible invariant finite-dimensional subspace.*
2. *M is an invariant linear subspace which is at most one-dimensional.*
3. *There is $\chi \in \Gamma^*$ and $f \in \ker(\chi - U)$, such that $M = \mathbb{C} \cdot f$.*

⁴Commutator subgroup and abelianisation?

⁵Meeting Notes: In the $\Gamma = \mathbb{Z}$ case, then H_{ds} can be broken up into one-dimensional objects that corresponds to an eigenfunction.

Definition 0.8. A unitary representation $U : \Gamma \rightarrow \mathcal{U}(H)$ of a topological group Γ is *strongly continuous* if $\Gamma \rightarrow H, x \mapsto U_x f$ is continuous for every $f \in H$.

Definition 0.9. For every discrete group Γ , the maps

$$L : \Gamma \rightarrow \mathcal{U}(L^2(\Gamma)), \quad x \mapsto L_x \quad (1)$$

$$R : \Gamma \rightarrow \mathcal{U}(L^2(\Gamma)), \quad x \mapsto R_x \quad (2)$$

with $L_x f := f \circ l_{x^{-1}}$ and $R_x f := f \circ r_{x^{-1}}$ for $f \in L^2(\Gamma)$ and $x \in \Gamma$ are strongly continuous unitary representations. We call L and R the *left* and *right regular representation* of Γ , respectively.⁶

Assume that $U : \Gamma \rightarrow \mathcal{U}(H)$ is a strongly continuous unitary representation of a discrete group Γ . Let further P be the orthogonal projection onto the fixed space $\text{fix}(U(\Gamma))$. Then

$$(Pf \mid g) = \lim_{N \rightarrow \infty} \frac{1}{|\Phi_N|} \sum_{\gamma \in \Phi_N} (U_\gamma f \mid g)$$

for all $f, g \in H$.

If $U : \Gamma \rightarrow \mathcal{U}(H)$ is a strongly continuous unitary representation of a compact group Γ , then U has discrete spectrum.⁷

Exercise 0.1. Show that

$$H = H_{\text{ds}} \oplus H_{\text{wm}}$$

and what happens to H_{ds} and H_{wm} when it is averaged by the Abstract Mean Ergodic Theorem for amenable groups.⁸

Proposition 0.2. Let (X, T) be a measure-preserving system. Then $L^2(X)_{\text{ds}}$ is an invariant Markov sublattice of $L^2(X)$.

Exercise 0.2 (Jamneshan and Kreidler, 2025, Lemma 6.1.12). Assume that $U : \Gamma \rightarrow \mathcal{U}(H)$ is a strongly continuous unitary representation of a discrete group Γ . Let further P be the orthogonal projection onto the fixed space $\text{fix}(U(\Gamma))$. Then $(Pf \mid g) = \lim_{N \rightarrow \infty} \frac{1}{|\Phi_N|} \sum_{\gamma \in \Phi_N} (U_\gamma f \mid g)$ for all $f, g \in H$.

Jamneshan, A. and Kreidler, H. (2025). *ISem 28: Ergodic structure theory and applications*.

⁶As Γ is discrete, then any map $\Gamma \rightarrow H$ is continuous for any topological space H . Thus, $\Gamma \rightarrow L^2(\Gamma)$ defined by $\gamma \mapsto L_\gamma f$ or $\gamma \mapsto R_\gamma f$ is always continuous. Hence, L and R are strongly continuous unitary representations.

⁷This Γ would be the compact group behind the factor subsystem of the larger system. For example, $\Gamma = \mathbb{Z}$ for the irrational α group rotation would give us compact $G = \{z \in \mathbb{C} \mid |z| = 1\}$. The homomorphism from $\Gamma \rightarrow G$ is $n \mapsto e^{2\pi i \alpha n}$.

⁸(Proposition 5.3.3)

$$H_{\text{wm}} = \left\{ f \in H \mid \lim_{N \rightarrow \infty} \frac{1}{|\Phi_N|} \sum_{\varphi \in \Phi_N} |(U_\varphi f \mid f)|^2 = 0 \right\}.$$